

The background features a light beige and cream color palette. On the left, there is a branch with several brown leaves. In the top right, there are small brown dots and a thin, elegant line. The bottom left and right corners also contain decorative elements, including a branch with leaves and abstract circular patterns with dots.

Capstone Presentation

Capstone Project 1

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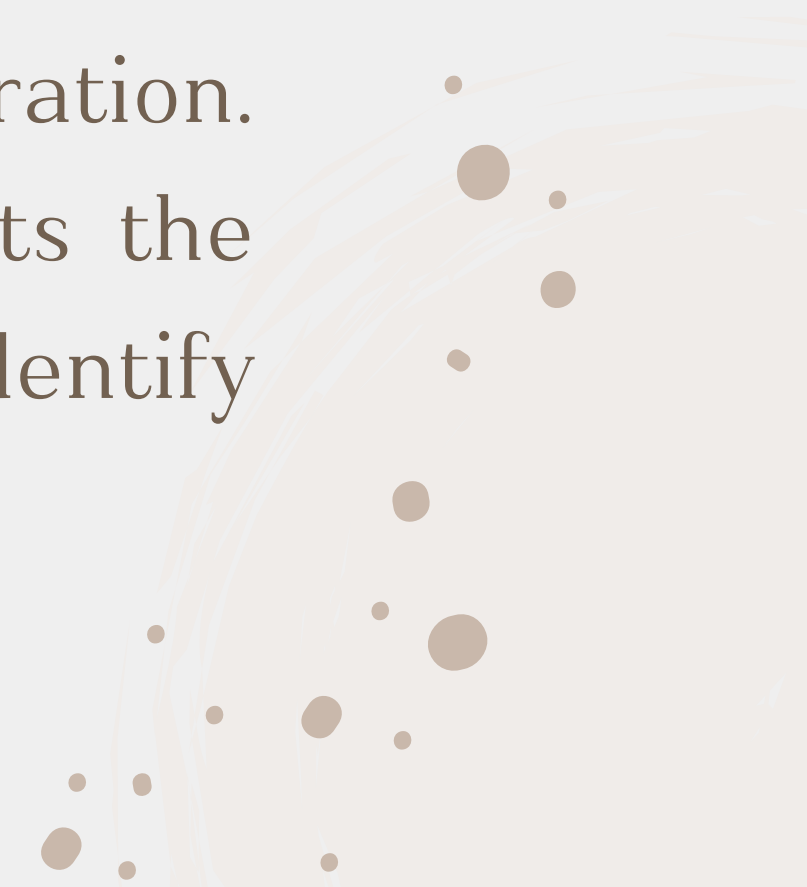
**A Web-Based Service
Management System with Data
Analytics using Machine Learning
Algorithms for Shoelotskey
Villamor – Pasay**



Research Problem

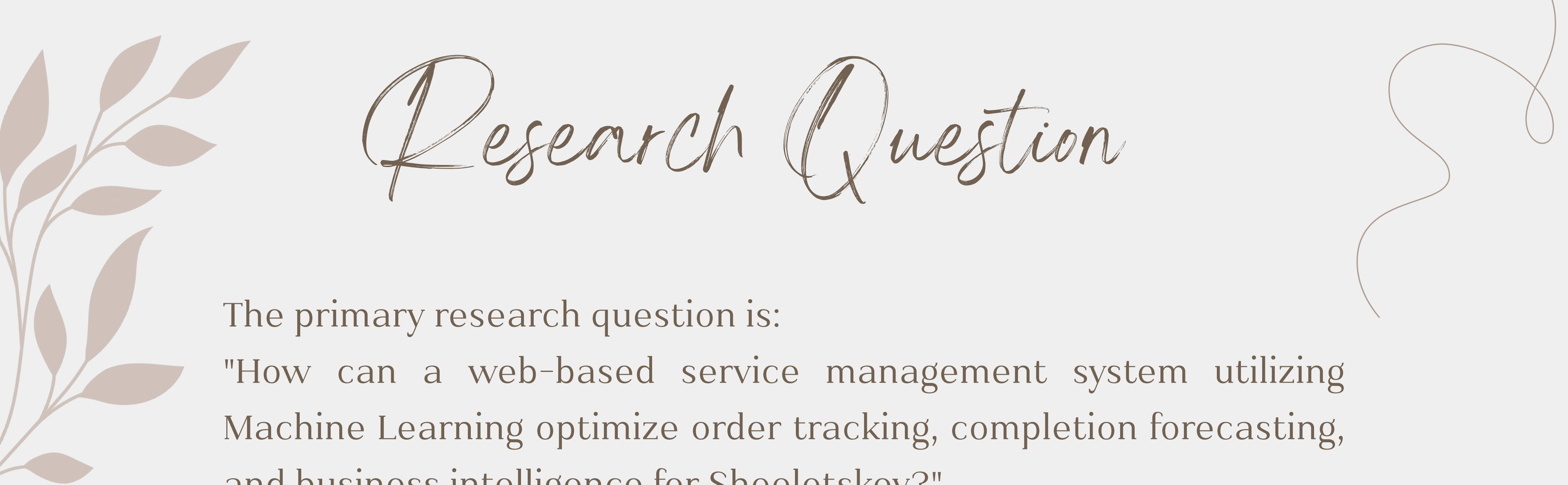


Currently, Shoelotskey operates using traditional, manual logbooks to track customer orders and shoe conditions. This manual approach is prone to human error, data loss, and slow retrieval of information. Furthermore, the business lacks a method for estimating service completion dates, often relying on mental guesswork that lead to missed deadlines or customer frustration. Finally, the absence of a centralized data system prevents the owner from seeing the big picture, making it difficult to identify which services are most profitable.



Problem Statement

Shoelotskey faces operational inefficiencies due to the reliance on paper-based record-keeping and a lack of data-driven decision-making tools. The current manual system fails to provide accurate service completion estimates, leading to inconsistent customer experiences. Additionally, the lack of an analytical dashboard prevents the management from identifying growth trends or operational bottlenecks. Without a web-based system integrated with Machine Learning, the business remains reactive rather than proactive, limiting its ability to scale and maintain high service standards with community and nearby cities.



Research Question

The primary research question is:

"How can a web-based service management system utilizing Machine Learning optimize order tracking, completion forecasting, and business intelligence for Shoelotskey?"

Specific Research Questions:

- How can a digital order entry be designed to effectively capture shoe details and service requirements compared to manual logbooks?
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Research Question

- What Machine Learning algorithm is most effective in predicting service completion dates based on historical turnaround data?
 - How can business analytics be visualized to provide the owner with actionable insights into financial and operational performance?
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Research Objectives

The primary goal is to develop the Shoelotskey Service Management System to streamline operations through automation and predictive modeling.

Specifically, the project aims:

- To develop a centralized order entry that digitizes customer information, shoe details, and requested services, replacing inefficient manual logbooks.
- To implement a Machine Learning model that predicts estimated service completion dates by analyzing historical turnaround times and current workload.



Research Objectives

- To generate a business analytics dashboard that provides the owner with real-time, data-driven insights into service volume, sales, and operational trends across various reporting periods (daily, weekly, monthly, annually).
 - To evaluate the system's effectiveness in terms of data accuracy, prediction reliability, and user satisfaction among Shoelotskey staff and management.
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Thank you!

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